2

ME1100 Thermodynamics

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Work

In **mechanics,** work is said to be done when **the point of application of a force moves through a distance in the same direction as the force**.

Work (W) = Force (F) * Distance (s)

Power is the rate of work done.

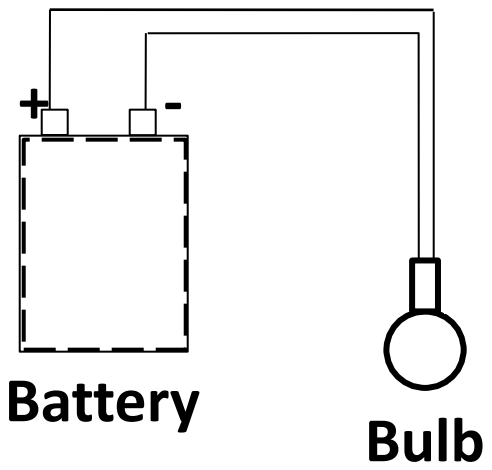
In many applications, the point of application or the distance moved cannot be precisely measured or not possible to define.

The thermodynamic definition of work, on the other hand, represents an energy interaction between the system and the surroundings.

Work is said to be done by a system, during a given operation, if the sole effect, external to the system could be reduced to the raising of weight through a distance.

Work

"raising of weight through a distance"

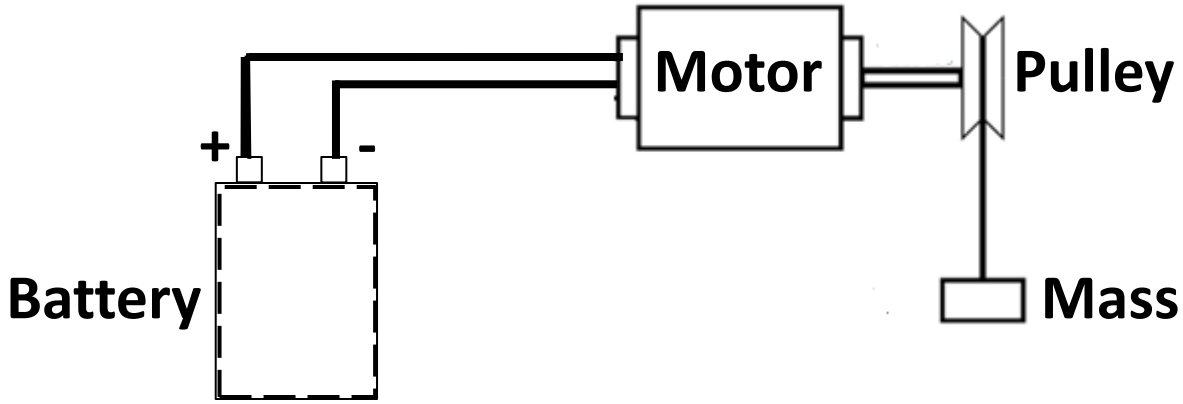


Let the voltage and current be 10 Volts and 5 Amps respectively.
If the battery is operated for a period of 5 minutes, the energy consumed during the operation of the battery can be calculated as:

energy consumed = voltage x current x time = 10 V x 5 A x 300 s = 15000 Joules

Considering the battery as the system, we can consider this interaction as a positive work of 15 kJ done by the battery on its surroundings (i.e. the bulb).

Where did the question of "*raising of weight through a distance*" come?



"could be reduced to the raising of weight through a distance"

Work

"sole effect"

When we replaced the bulb with a motor connected to a mass (through a rope tied to a pulley and then connected to the motor shaft), the process undergone by the system (battery in this case) was left untouched.

Knowing motor and mass information and assuming frictionless pulley and lossless motor, we can find the rise of the mass.

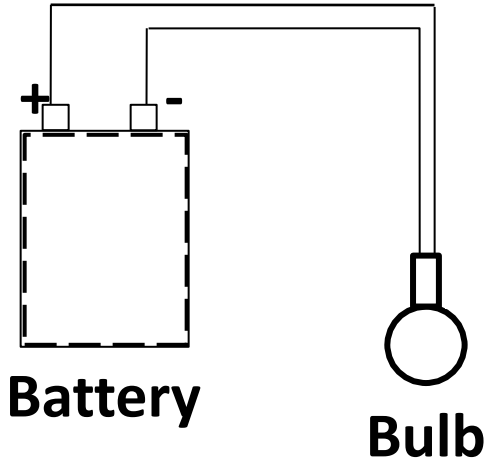
If there are any frictional losses and electrical losses in the motor, they will result in heat dissipation; then, apart from the lifting of the weight, there will also be slight changes in the temperature of the surrounding objects.

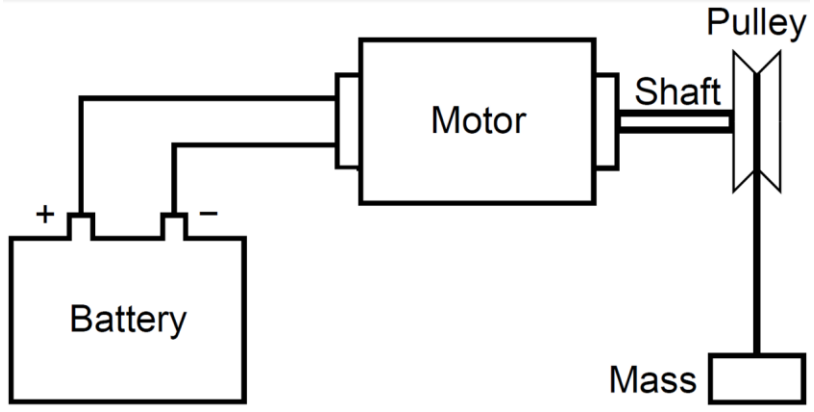
Hence, the height that can be raised (or, the work done on the mass) will be different from the work done by the battery.

So the criterion of "sole effect" becomes important. Fortunately, achieving nearly lossless motor or nearly frictionless pulley is achievable.

Work

- **Therefore, we need to state this clearly by using words like "*by*" or "*on*".**
- **Since work is an energy interaction between the system and the surrounding, there must be a donor and a receiver.**
- **When a system transfers energy in the form of work, it does positive work.**
- **At the same time, the surroundings will do an equal amount of negative work.**





If all components are included in the system, there is no work interaction.

For battery alone as a system, clearly it does work by lifting the mass.

Work done BY the battery: positive (15 kJ)

Work done ON the bulb: negative (15 kJ)

Work done by the battery + work done on the bulb =0

For motor alone as a system, it receives electrical energy (work) from the battery and rotates the shaft to lift the weight.

For mass alone, work is negative.

$$W_{sys,1} + W_{sys,2} + W_{sys,3} + \ldots\ldots + W_{sys,n} = 0$$

Work: importance of resistance offered

- **A very important aspect of work interaction is that the process carried out by the system must be resisted by the surroundings.**
- **If the surroundings do not offer any resistance, then the work done by the system will become zero.**

A body of mass m falls from an elevation h to the ground in the vacuum. What is the work done by the body on the surroundings?

Possible answers: (a) mgh (b) –mgh (c) zero

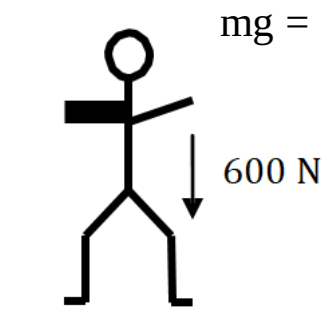
Correct answer : zero. Why ?

When the body falls through the vacuum, it is not resisted by the surroundings (vacuum). Hence work done is zero.

There is merely a conversion of potential energy into kinetic energy but no energy crosses the system boundary.

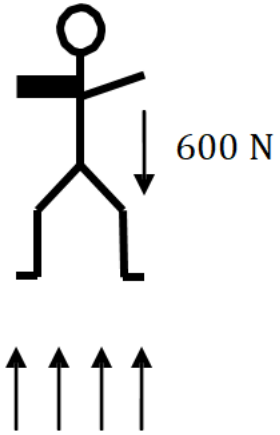
If vacuum is replaced by air, the work done by the body on the surroundings: air resistance multiplied by h.

Work: importance of resistance offered



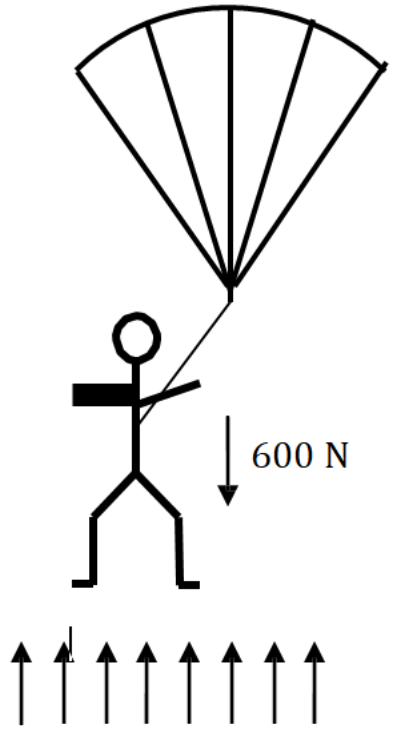
Air resistance
= zero (vacuum)

(a) unresisted



Air resistance
= 50 N

(b) partially resisted



Air resistance
= 600 N

(c) fully resisted

Work interaction between system (person) and surroundings

0

50 N x distance

600 N x distance

Work-energy theorem from high school mechanics

The work-energy theorem states that **the net work done by the forces on an object equals the change in its kinetic energy**

Let us start with the definition of work: $dW = \overrightarrow{F} \cdot dz \Rightarrow W = \int F dz$

If we consider a conservative force (like gravitational force), then we can write force in terms of its potential (*U*)

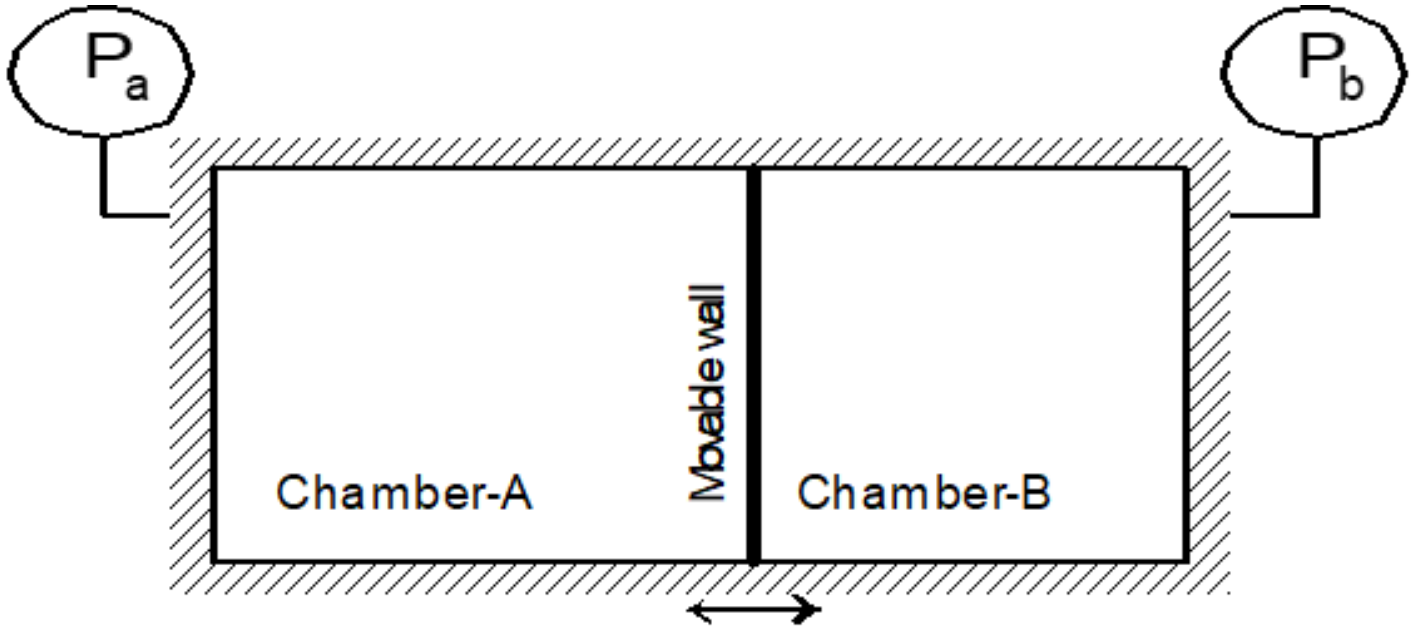
$$\vec{F} = -\frac{dU}{dz} \Rightarrow W = \int \left(-\frac{dU}{dz}\right) dz = -\Delta U$$

Thus, by the application of work-energy theorem (in the absence of any other force, like the frictional force), we get

$$\mathrm{W} = -\Delta U = \Delta KE \Rightarrow \Delta KE + \Delta U = 0$$

Hence, the work-energy theorem suggests that gravitational force leads to the conservation of mechanical energy of the object (we are using the term thermodynamic system)

What is the value of work?



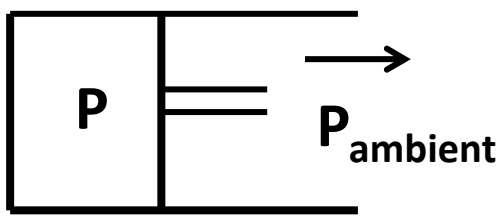
If the pressure in the two chambers is not the same (Pa <> Pb) the wall moves till the pressures in the two chambers become equal.

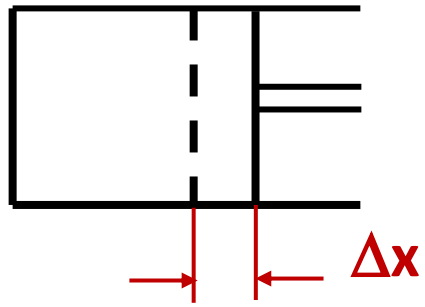
What is the work done in the process?

ZERO

Since there is no work interaction at the system boundary, the thermodynamic work interaction is zero in this case, even though some work is done by the gases on the movable wall.

Displacement work: piston moves a distance Δ x





$$P = P_{ambient} + \Delta p$$

- There are three separate objects to consider:
  - Gas enclosed inside the cylinder
  - Piston
  - Ambient air

The work interaction during the process causing a change of state from 1 to 2 is:

$$W_{1\text{-}2} = \int \delta W = \int p dV$$

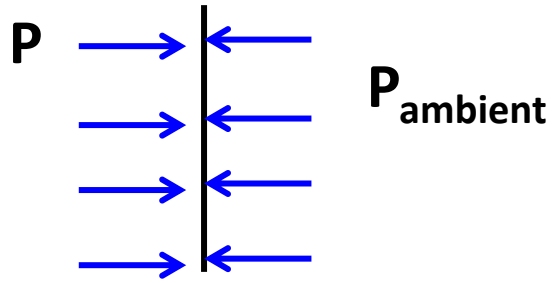
The process that relates p and V should be known to calculate **the displacement work**

Displacement work: piston moves a distance Δ x

- Consider the gas inside the cylinder as the system. Everything else is surrounding
- Work interaction between the gas and the surroundings:

$$W_1 = P * A * \Delta x$$

- Consider the piston as the system.



Net work interaction between piston and surroundings:

W_2= work done by the piston on the surrounding + work done by the gas on the piston

(+ve) (-ve)

$$W_2 = (P_{ambient} - P) * A * \Delta x$$

Displacement work: piston moves a distance Δ x

Consider the atmospheric air as the system.

Work interaction between atmospheric air and surroundings:

$$W_3 = -P_{ambient} * A * \Delta x$$

Total work interaction:

$$W = W_1 + W_2 + W_3$$

$$= (P * A * \Delta x) + \{(P_{ambient} - P) * A * \Delta x\} + (-P_{ambient} * A * \Delta x)$$

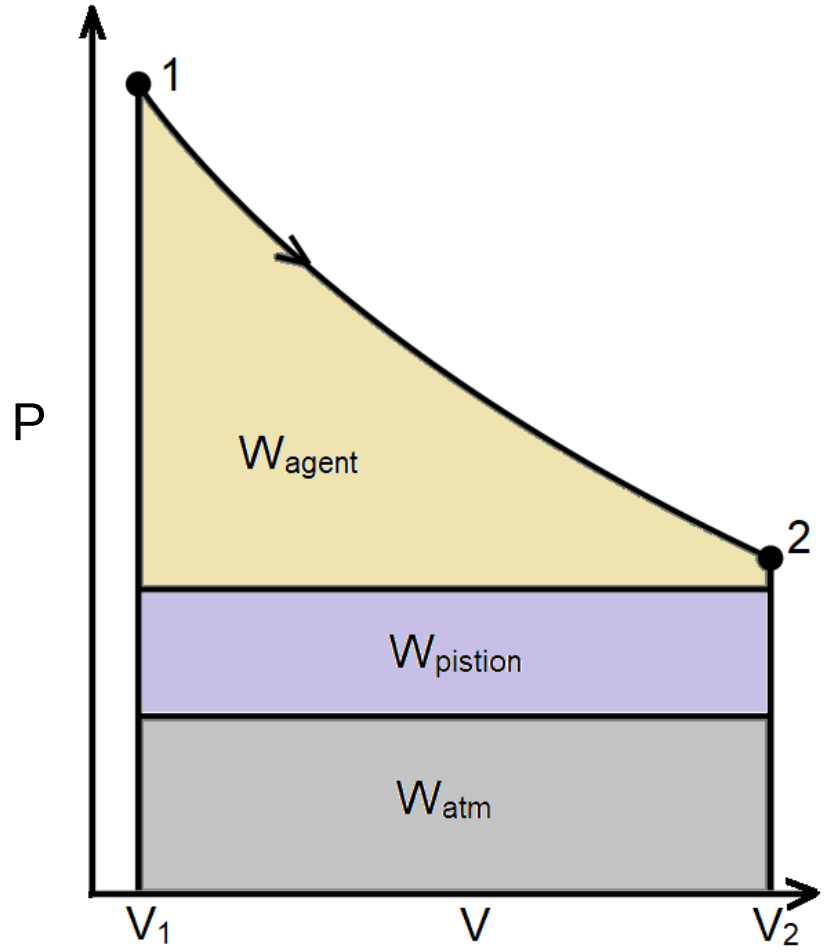
$$= 0$$

Displacement work: piston moves a distance Δ x

Consider a piston-cylinder arrangement with a working substance initially at pressure P_1 and occupying a volume V_1.

An external agent initially provides a resisting force F_1 that gradually diminishes causing the system to undergo an expansion process to a final state labelled 2.

Work interactions for atmosphere, piston and the agent are shown in the p-V diagram.



Displacement work done during different thermodynamic processes

- **Constant pressure process**

$$W_{1-2} = \int_1^2 P\, dV = P(V_2 - V_1)$$

- **Isothermal process, where PV = constant = C**

$$W_{1-2} = \int_1^2 P\, dV = \int_1^2 \frac{C}{V} dV = PV \ln(V_2/V_1)$$

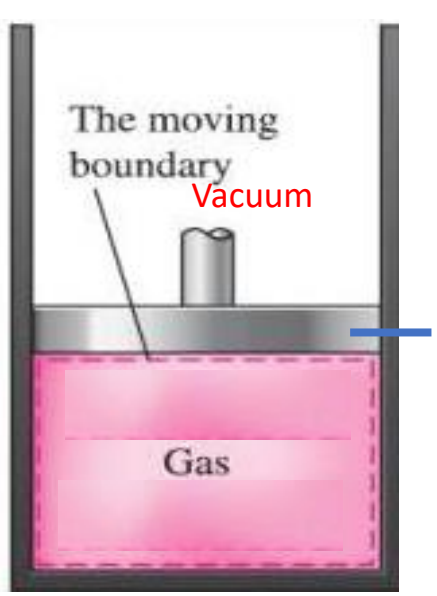
- **Process where PV^n = constant = C**

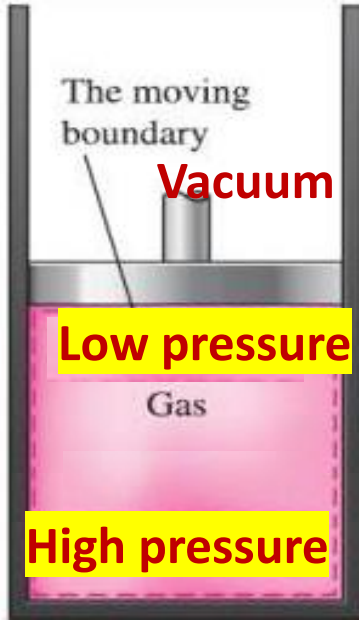
$$W_{1-2} = \int_1^2 P\, dV = \int_1^2 \frac{C}{V^n} dV = \frac{C(V_2^{1-n} - V_1^{1-n})}{1-n} = \frac{P_1V_1 - P_2V_2}{n-1}$$

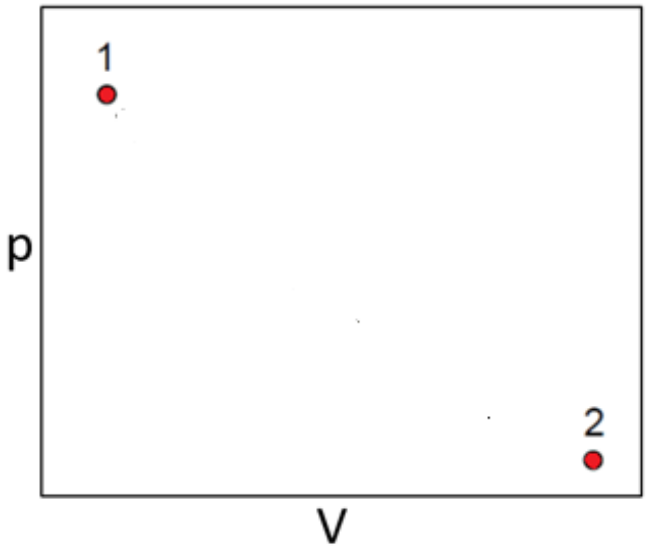
- **Constant volume process**

$$W_{1-2} = 0$$ **since *dV*=0**

Unrestrained Expansion







When **suddenly released**, the **piston** experiences a **very large acceleration**

The **external pressure** exerted by the surroundings **is zero** **– no work on surroundings: ZERO WORK DONE**

The **pressure within the gas** is **not uniform**

High pressure at the bottom end and **low pressure adjacent to the piston** due to rapid acceleration

Loss of internal equilibrium

Intermediate state identification not possible – path not known

Other forms of work

- **Spring work**

$$W_{spring} = \frac{1}{2}kx^2$$

(k = spring constant and x is the displacement)

- **Electrical work of a capacitor**

$$W_{capacitor} = \frac{1}{2}CV^2$$

(C= C is capacitance and V is voltage)

- **Electrical work of a resistor**

$$W_{resistor} = I^2Rt$$
$$W_{resistor} = VIt$$

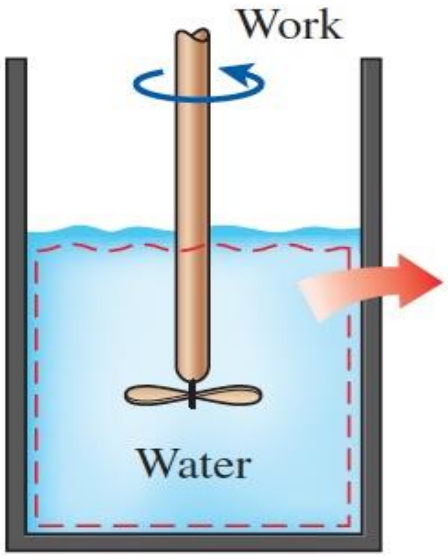
(where I = current, R = resistance V = Voltage, t = time)

Other forms of work

- **Shaft work**

$$W_{shaft} = 2\,\pi\,N\,\tau\,t$$

(where τ = Torque, N is rps, t=time)



- **Lifting of a mass m by height h (gravitational work)**

$$W_{gravity} = mgh$$

Remember this is not the same as that when the mass falls by the same distance

- **Paddlewheel work**
  - **Dissipative**
  - **Friction is required**

Word of caution: The sign of work depends on whether the system receives energy and gives out energy

Thank You